1 FIG-215 — From-scratch failure snapshot atlas

Group: MD-FROM-SCRATCH-FAILURE-ATTRIBUTION **Evidence chain:** manifested **Status:** snapshot atlas for inspected configurations; no accepted transport result

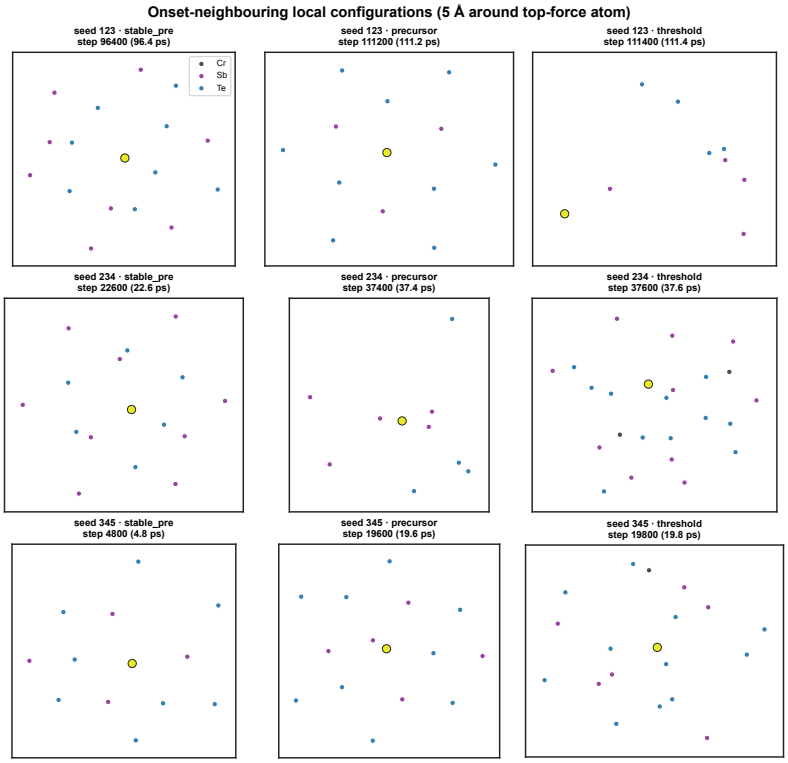


Figure 1: A–I, representative local configurations retained from the original and reproduction branches at the selected diagnostic windows. Each tile preserves the branch and seed identity used in the failure-attribution audit. The atlas is an inspection aid for follow-up calculations, not a structural assignment or accepted transport result.

Results text draft. The snapshot atlas retains nine branch- and seed-resolved configurations spanning the audited campaign. These configurations provide concrete follow-up targets while preserving the distinction between observed local structure and an inferred failure mechanism.

Discussion and boundary. The snapshots are representative diagnostic evidence only. They do not certify a local coordination change, identify the microscopic cause of instability, or substitute for independent DFT validation. No transport value is assigned to an unstable branch from this atlas.

Article placement. Proposed article location: Supplementary Information / from-scratch instability and failure-attribution audit; not yet a main-text structural-mechanism claim.

Reviewer linkage: overall status PENDING.

- **REV-C9 (PENDING)**: Inspect whether representative local configurations support the proposed from-scratch failure follow-up. *Completed:* Nine representative local configurations are retained with branch and seed identity. *Still required:* Validate local-structure interpretation against independent reference calculations before making a mechanistic claim.

Evidence pointers.

- Data/processed snapshots: `/Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/results_all_replicates/snapshots`
- Run ledger: `/Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/results_all_replicates/run_summary.csv`
- Plot/processing script: `/Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/plot_failure_attribution_all.py`

- Plot Atlas and figure manifest: /Users/alina/Documents/Codex/plot_library; /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/figures/figure_manifest.json
- Asset SHA256: 80d3a1072d4a19e3fcac9d7881d65a6a276a3be659a85a864373346df1e98446

Machine-collected evidence candidates (generated from the analysis manifest; do not edit this file).

- **Diagnostic report:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/results_all_replicates/RESULTS_DISCUSSION.md
- **Raw or processed data:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/results_all_replicates/diagnostic_manifest.json
- **Plot or processing script:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/plot_failure_attribution_all.py
- **Plot Atlas source:** /Users/alina/Documents/Codex/plot_library

Audit gap / next action: Use the retained snapshots to define an independent structural validation set; do not promote visual inspection to a local-mechanism conclusion without reference calculations.